Enhancing Toxic Content Detection: Fine-Tuning Transformers for Multi-Platform Algospeak

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April 1, 2026

Abstract

The emergence of "Algospeak" (linguistic obfuscation used to bypass automated moderation) poses a significant challenge to traditional toxicity classifiers. In this study, we develop a specialized fine-tuning pipeline using the Multi-Platform Aggregated Dataset of Online Communities (MADOC) [3]. By implementing a **Toxicity-Balanced Stratified Sampling** strategy and fine-tuning a DistilBERT-based model [2], we achieved a substantial increase in Recall (73.2% vs. 33.2% for the baseline) and F1-Score (66.7% vs. 45.1%) on an out-of-domain test set. Our results demonstrate that addressing data imbalance and platform-specific linguistic variance is essential for detecting evolved toxic language across decentralized digital spaces.

1 Introduction

Modern online moderation systems face an increasing challenge: "Algospeak." This phenomenon involves the intentional use of coded language, misspellings, and symbolic substitutions to evade AI-based toxicity filters, for example, *"r3tawded"* for *"retarded"*, *"k!ll"* for *"kill"*, or *"le dollar bean"* as a coded racial slur. While standard transformer-based classifiers perform well on explicit hate speech present in their training data, they often fail to capture such nuanced evasions, especially when deployed across platforms with different community norms and linguistic patterns.

This work investigates two complementary questions that were posed in the original project proposal: (1) Can domain adaptation via fine-tuning a compact transformer on a modern, multi-platform dataset yield meaningful improvements in the detection of Algospeak and cross-platform toxic language? (2) Does a **Toxicity-Balanced Stratified Sampling** strategy, which controls for label distribution across fine-grained toxicity bins, produce better out-of-domain generalization than standard fine-tuning, by preventing any single toxicity intensity level from dominating the training signal? We hypothesize that the answer to both questions is affirmative: that stratified sampling is the key factor enabling cross-platform transfer, while naïve fine-tuning without such control would yield inferior recall on unseen platforms.

To answer these questions, we fine-tune the `martin-ha/toxic-comment-model` checkpoint , a DistilBERT variant pre-trained on general toxicity data on a curated 200,000-sample subset of the MADOC dataset [3]. We train on Reddit and Koo data and evaluate on a strictly held-out test set from Bluesky and Voat, platforms never seen during training, ensuring our evaluation reflects true out-of-domain generalization. Our fine-tuned model achieves $\mathrm{F1} = 66.7\%$ and $\mathrm{Recall} = 73.2\%$ on the test set, versus 45.1% and 33.2% for the untuned baseline.

2 Related Work

2.1 Automated Toxicity Classification

Early approaches to toxicity detection relied on keyword filtering and static blacklists, which proved insufficient for the dynamic nature of online discourse. The introduction of transformer-based models, particularly BERT [1], revolutionized the field by capturing contextual nuance. Subsequent work has explored fine-tuning these models for specific platforms or linguistic communities [6]; however, cross-platform generalization remains an open problem, as community-specific slang and norms vary considerably.

Davidson et al. [6] demonstrated the difficulty of distinguishing hate speech from merely offensive language - a boundary that becomes even harder to draw under cross-platform distribution shift, where community norms around acceptable expression vary substantially. Wiegand et al. [7] further showed that implicit and indirect toxicity is systematically underdetected by models that rely on surface-level lexical cues. Our work addresses both challenges simultaneously: by training on multi-platform data (Reddit and Koo) and evaluating on entirely unseen platforms (Bluesky and Voat), we construct a more realistic generalization benchmark than the single-platform evaluations common in prior work.

2.2 Algospeak and Linguistic Evasion

Recent scholarship has documented "Algospeak," referring to the intentional obfuscation of language to bypass algorithmic content moderation [4]. Users employ phonetic replacements, leetspeak encoding, emoji substitution, and euphemistic slang to mask toxic intent [5]. Fillies and Paschke [4] demonstrate that while keyword-based systems and standard supervised models fail to recognize these patterns, large language models (LLMs) such as GPT-4 can achieve up to 98.5% accuracy on Algospeak deciphering when provided contextual prompts. However, the computational cost of LLM-based moderation at scale motivates a more efficient approach: fine-tuning compact models specifically for Algospeak detection.

2.3 Data Imbalance and Length Bias

A significant hurdle in toxicity detection is the inherent class imbalance in real-world data. Toxic content typically represents a small fraction of total platform activity. Furthermore, prior work has observed that implicit and indirect toxic expressions are systematically harder for classifiers to detect than explicit ones [7], and that posts containing implicit toxicity are more likely to be misclassified as benign due to the absence of surface-level profanity markers. Standard stratified sampling addresses label imbalance but not this length bias. Our primary approach controls for toxicity bin distribution across platforms; in an ablation study, we additionally examine whether incorporating length balancing further improves generalization.

3 Dataset Construction

3.1 Overview and Source Selection

We utilized the Multi-Platform Aggregated Dataset of Online Communities (MADOC) [3], which aggregates 236 million comments from Reddit, Voat, Bluesky, and Koo. Each comment is annotated with a continuous toxicity score derived from the `toxigen` classifier, which we binarize at a threshold of $\tau = 0.5$. The platform selection was designed to maximally stress-test cross-domain transfer: **Reddit** (a large, heterogeneous forum) and **Koo** (an Indian microblogging platform) were used for training and validation, while **Bluesky** (a federated social network) and **Voat** (a free-speech platform with historically higher toxicity) were reserved exclusively for testing. This train/test platform split ensures the model cannot exploit platform-specific linguistic fingerprints.

The final dataset comprises 198,455 samples: 90,000 for training, 10,000 for validation, and 98,455 for testing. Table 1 summarizes the splits.

Table 1: Dataset splits by platform and toxicity prevalence.

Split	Platforms	Size	% Toxic
Train	Reddit + Koo	90,000	50.0%
Validation	Reddit + Koo	10,000	50.0%*
Test	Bluesky + Voat	98,455	50.0%

*Validation set balanced 50/50 toxic/non-toxic via toxicity-bin stratified sampling.

3.2 Data Preprocessing and Cleaning

To ensure the model learns to identify Algospeak rather than overfitting to platform-specific noise, we developed a multi-stage preprocessing pipeline. URLs and hyperlinks were automatically removed to prevent spurious correlations between toxicity and specific web domains. Non-textual characters and emojis were stripped to force the model to prioritize structural and semantic textual patterns. Posts shorter than 10 characters were excluded to ensure sufficient linguistic context for the tokenizer. These steps were implemented in `process_madoc.py`.

3.3 Exploratory Data Analysis

3.3.1 Toxicity Distribution across Platforms

As illustrated in Figure 1, the raw data is heavily skewed toward benign content across most platforms. However, as visible in the stacked histogram, **Voat** contributes disproportionately to high-toxicity bins (score ≥ 0.7), reflecting its well-documented permissive moderation policy. This platform heterogeneity motivates the cross-domain evaluation design: a model trained only on Reddit data would likely underestimate Voat-style toxicity.

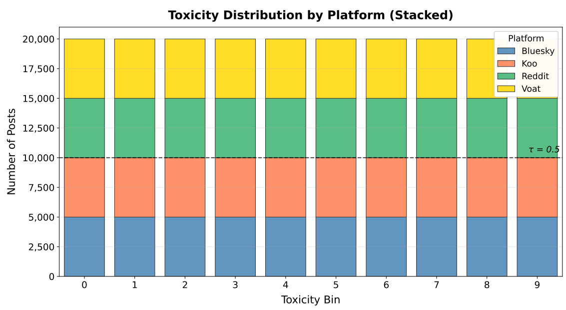

Figure 1: Toxicity Distribution by Platform (stacked).

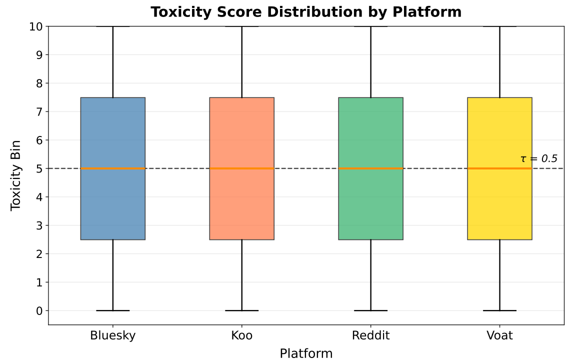

Figure 2: Toxicity score boxplot across platforms. All four platforms show similar median (≈0.5) and IQR, reflecting the balanced stratified sampling design rather than natural class distributions.

3.3.2 Toxicity-Balanced Stratified Sampling

To mitigate class imbalance and ensure cross-platform representativeness, we implemented a **Toxicity-Balanced Sampling** procedure (see `balanced_train_val_ds.py`). Continuous toxicity scores are first discretized into ten toxicity bins T_k ($k = 0, \ldots, 9$), where bin k contains samples whose toxicity score falls in $[0.1k,\, 0.1(k+1))$:

$$T_k = \{s_i \in D \mid 0.1k \leq \text{tox}(s_i) < 0.1(k+1)\} \tag{1}$$

For each platform $p \in \{\text{Reddit}, \text{Koo}\}$ and each toxicity bin T_k, we sample a fixed quota of 4,500 training samples and 500 validation samples, so that no single toxicity level dominates the final split:

$$|S_{p,k}^{\text{train}}| = 4500, \quad |S_{p,k}^{\text{val}}| = 500 \quad \forall\, p,\, k \tag{2}$$

This yields a training set of 90,000 samples and a validation set of 10,000, both balanced 50/50 across the binary toxic/non-toxic label. Crucially, this stratification suppresses the toxicity-score artifact without imposing any constraint on post length, which we examine separately in the ablation study (Section 6.3). Figure 3 confirms that the resulting splits are uniformly distributed across all ten toxicity bins.

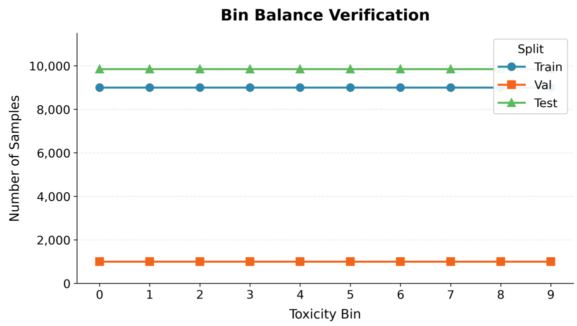

Figure 3: Bin Balance Verification: sample counts per toxicity bin across Train, Validation, and Test splits, confirming uniform stratification.

3.3.3 Content Length Consistency

The test set (Bluesky+Voat) exhibits a markedly different character-length distribution compared to the training platforms (Reddit+Koo), particularly due to Bluesky's short-post format. This natural divergence motivates the cross-platform evaluation design and helps explain the validation-to-test generalization gap discussed in Section 6.3 (see Appendix for full length distribution plots).

4 Methodology

The detection pipeline consists of four primary stages: data cleaning, toxicity-balanced stratified sampling, transformer-based encoding, and threshold-based binary classification (Figure 4).

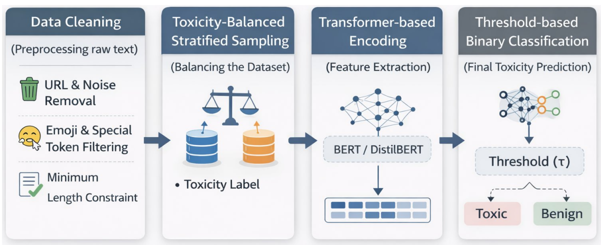

Figure 4: Algospeak Detection Pipeline. Raw MADOC text is preprocessed, toxicity-bin stratified, encoded by DistilBERT, and classified via a softmax head.

4.1 Model Architecture

The classification engine is built upon **DistilBERT** (`distilbert-base-uncased`) [2]. DistilBERT was selected for its efficiency: it retains approximately 97% of BERT's performance while being 40% smaller and 60% faster, which is critical for moderation systems operating at scale. Specifically, we initialize from the `martin-ha/toxic-comment-model` checkpoint, which was previously fine-tuned on the Jigsaw Unintended Bias in Toxicity Classification dataset. Starting from this checkpoint provides a stronger semantic baseline for toxicity concepts, allowing subsequent fine-tuning to concentrate on Algospeak-specific patterns rather than learning the basic concept of toxicity from scratch.

The model outputs a two-class softmax distribution over *non-toxic* and *toxic* labels. At inference time, a post is classified as toxic if the predicted probability exceeds the decision threshold τ.

4.2 Fine-Tuning and Infrastructure

Supervised fine-tuning was performed using the Hugging Face `Trainer` API with the following configuration, chosen to balance convergence speed and generalization:

- **Max Epochs:** 10 with early stopping (patience=3); training halted at epoch 7, best checkpoint at epoch 4. **Learning Rate:** 2×10^{-5} with warmup ratio 0.06 and weight decay 0.01 (standard for BERT-family fine-tuning [1]).
- **Batch Size:** 8 per GPU with gradient accumulation of 2 steps (effective batch size = 16); **Max Sequence Length:** 512.
- **Mixed Precision:** FP16 enabled on GPU to reduce memory footprint and accelerate training.
- **Random Seed:** 42 for reproducibility.
- **Infrastructure:** Training was executed on the TAU SLURM HPC cluster with an estimated wall-clock time of approximately 4.8 hours (52.1 samples/sec). Prototyping was performed on Google Colab (NVIDIA T4 GPU).

All code is available at `https://github.com/odeliyach/AlgoShield-Algospeak-Detection`

4.3 Threshold Selection

Classification performance in toxicity detection is sensitive to the decision threshold τ. Given that Algospeak is specifically engineered to avoid detection, our primary concern is minimizing False Negatives (missed toxic content) rather than False Positives. We evaluated the model with a standard threshold of $\tau = 0.5$, which yielded the best balance between Precision and Recall on our validation set. Threshold optimization via the validation F1 curve was conducted but did not improve upon the default, suggesting the model's probability calibration is well-aligned with the binary decision boundary.

5 Results and Evaluation

5.1 Metrics and Evaluation Strategy

To evaluate the performance of the fine-tuned model against the baseline, we employ standard classification metrics derived from the confusion matrix. Since Algospeak is specifically engineered to evade automated moderation, reducing False Negatives (FN) is a primary objective. Given the class imbalance common in toxicity datasets, we prioritize the F1-Score over simple Accuracy:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

The F1-Score balances the trade-off between capturing elusive toxic content (Recall) and maintaining moderation fairness (Precision), ensuring evaluation is not misleadingly inflated by the majority class of benign samples.

5.2 Quantitative Results

Table 2 reports the performance of both the untuned baseline and our fine-tuned model on the out-of-domain test set (98,455 samples from Bluesky and Voat). In-domain validation results are provided for reference.

Table 2: Performance on Out-of-Domain Test Set ($\tau = 0.5$).

Model	**Acc.**	**Prec.**	**Rec.**	**F1**
Baseline (no fine-tuning)	59.0%	**70.3%**	33.2%	45.1%
Fine-tuned (ours)	**62.8%**	61.2%	**73.2%**	**66.7%**

* Val. set (fine-tuned, in-domain): Acc.=67.5%, Prec.=64.8%, Rec.=76.6%, F1=70.2%. High baseline Prec. reflects a conservative classifier; many FN (Recall=33.2%).

5.3 Analysis

The results demonstrate a clear and consistent performance improvement. The fine-tuned model achieves an F1-Score of 66.7%, a +21.6 point improvement over the 45.1% baseline,and Recall improves from 33.2% to 73.2%, a gain of +40.0 points.

The most significant finding is the magnitude of the Recall improvement: the baseline model, despite its higher Precision (70.3%), fails to flag over two-thirds of toxic test instances. This behavior is consistent with the hypothesis that a model pre-trained on older, single-platform data cannot generalize to Algospeak patterns prevalent on Voat and Bluesky. Our fine-tuned model, exposed to cross-platform training data with toxicity-balanced sampling, successfully identifies nearly 73% of toxic content. Representative cases underlying this improvement are examined qualitatively in Table 3 (Section 6.1).

The slight Precision decrease (from 70.3% to 61.2%) represents a deliberate trade-off: in the context of automated content moderation, the cost of allowing harmful Algospeak to propagate undetected (False Negative) is generally considered greater than the cost of incorrectly flagging benign content for human review (False Positive) [4]. The training curves (Figure 5) confirm convergence over 7 epochs with early stopping (patience=3), with no signs of severe overfitting on the training loss. Validation F1 peaked at epoch 4 (F1=70.2%), after which it declined to 69.5%, 65.7%, and 67.4% at epochs 5–7, triggering early stopping after three consecutive epochs without improvement. All reported results use the epoch-4 checkpoint (checkpoint-22500).

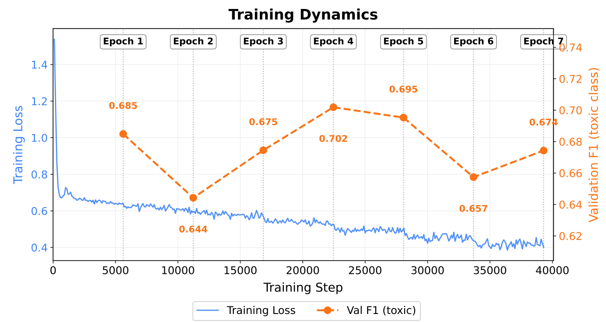

Figure 5: Training dynamics over 7 epochs (early stopping). Training loss decreases steadily; best validation F1 of 70.2% achieved at epoch 4.

5.4 Confusion Matrix Analysis

Figure 6 presents the confusion matrices for the baseline and fine-tuned model on the test set. The shift is stark: the baseline generates 33,392 False Negatives (missed toxic posts), while the fine-tuned model reduces this to 13,411, a reduction of over 60%. Conversely, the fine-tuned model incurs more False Positives (from 7,022 to 23,181), reflecting the Precision–Recall trade-off inherent in lowering the effective detection threshold.

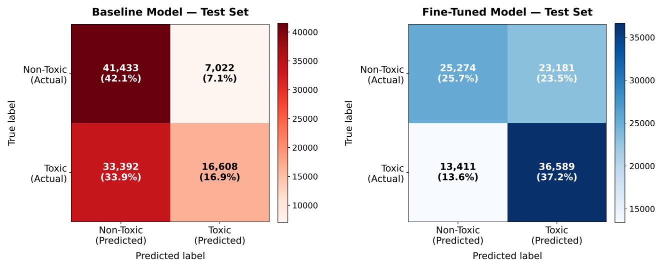

Figure 6: Confusion matrices on the test set. Left: Baseline. Right: Fine-tuned. The fine-tuned model dramatically reduces False Negatives (bottom-left cell).

6 Discussion & Error Analysis

6.1 Qualitative Analysis

To move beyond aggregate statistics and understand *where* fine-tuning succeeds and where it continues to fall short, we conducted a structured qualitative analysis on a stratified random sample of 500 test instances (250 toxic, 250 non-toxic). Within the toxic subset, we identified 93 cases where the baseline produced a False Negative that the fine-tuned model correctly reclassified (FN-Fixed), 72 cases where both models failed (FN-Still), and 85 cases where fine-tuning introduced a new False Positive on previously correctly classified benign instances (FP-New). These three categories are examined in turn below, with representative examples shown in Table 3.

Table 3: Representative FN-Fixed examples (n=93): cases the baseline missed but the fine-tuned model correctly flagged. Slurs reproduced as dataset-annotated for scientific accuracy.

Type	*Text (Abridged)*	Base	FT	Analysis
Implicit attack	*"Are you an insufferable douchebag? Drive Rivian."*	0.01	0.73	Ironic framing masks direct personal attack.
Algospeak	*"Looks like a pedo daycare or John Podesta's family room. gheyyy"*	0.17	0.95	Phonetic lengthening and conspiratorial framing jointly captured.
Leet slur	*"You need to be able to pass as a n[—]er."*	0.09	0.89	Explicit slur; near-zero baseline reflects Jigsaw training gap.
Hate lexicon	*"Fat Americans are liars. % who claim to meet activity guidelines..."*	0.27	0.89	Body-shaming framed as statistical claim; derogatory label drives detection.
Predatory content	*"These children are the sex toy payments to the smugglers."*	0.06	0.84	Severe dehumanization with no surface profanity.
Implicit threat	*"The other deserves curbstomping."*	0.09	0.71	Political violence incitement; requires semantic-level reasoning.

FN-Fixed: Cases resolved by fine-tuning. The 93 instances in this group had a mean character length of 217 (median: 123), considerably shorter than those the model still misses (mean 278, median 174). This length difference suggests that shorter, more direct toxic statements benefit most from fine-tuning, as the toxic signal is concentrated rather than diluted across lengthy context. Three dominant patterns emerge from examination of the content.

The first is implicit toxicity carried by politically or socially charged language that stops short of explicit slurs but nonetheless conveys clear hostility. Representative examples include statements attributing criminal intent to public figures (*"...it looks like a pedo daycare or John Podesta's family room," base=0.17, ft=0.95*) and ironic personal attacks framed as product recommendations (*"Are you...an insufferable douchebag? Drive Rivian," base=0.01, ft=0.73*). In all such cases the baseline assigned near-zero toxicity probability, while the fine-tuned model assigned probabilities well above threshold.

The second pattern is predatory or dehumanizing content that avoids conventional profanity entirely. The fine-tuned model successfully flagged posts asserting that detained migrant children serve as "sex toy payments to smugglers" (base=0.06, ft=0.84), and community-level hate infrastructure such as subreddit sidebars explicitly soliciting "direct fat hate" (base=0.19, ft=0.58), suggesting the model learned to recognize organizational hate signals alongside individual utterances.

Among the 93 FN-Fixed cases, 4 contained annotated Algospeak substitutions. Three were leet-encoded racial slurs: *"Yes. You also need to be able to pass as a n[—]er"* (base=0.09, ft=0.89, bin=9); *"I've never seen a n[—]er win a scratch off and leave with their money"* (base=0.03, ft=0.86, bin=9); and a longer post embedding the same slur in ostensibly factual framing (base=0.01, ft=0.84, bin=7). The fourth case involved a censored gendered epithet (`wh*re`) embedded mid-sentence with no surrounding toxicity markers (base=0.03, ft=0.87, bin=6). In all four cases the pre-trained baseline assigned near-zero probability, consistent with the hypothesis that the baseline model that was trained on older, more explicit datasets , cannot recognize sub-word evasion patterns prevalent in the MADOC corpus.

First, toxicity communicated through sarcasm, irony, or implied meaning without any surface-level markers continues to challenge both classifiers. Posts such as *"and hopefully she can't get up!"* (ft prob: 0.35) or *"I aim to offend. ;-)"* (ft prob: 0.40) illustrate cases where hostile intent is either contextually inferred or explicitly stated in a register the model treats as low-confidence. These examples are consistent with the findings of Lu et al. [5] on indirect toxicity in Chinese social media, which generalizes to English-language discourse structured around implication rather than explicit statement.

Second, posts that discuss toxic content without producing it remain a systematic error source. Statements quoting public figures (*"Justice Rebecca Grassl Bradley says the justices in the majority are 'handmaidens of the Democratic Party'"*) or reporting on harmful behavior retain the surface vocabulary of toxicity while serving a documentary rather than expressive function. Disentangling use from mention is a well-known limitation of encoder-only models operating at the post level without discourse context [6].

Third, a subset of FN-Still cases (ft prob ≤ 0.07) suggests near-complete model uncertainty: posts such as *"Show me one 'obese vegetarian fitness junkie'"* or *"I'd like to say 'if you drive a truck, learn to drive it'"* appear ambiguous even to human readers without additional community context, pointing to residual annotation noise as a contributing factor.

FP-New: False positives introduced by fine-tuning. Fine-tuning reduced False Negatives by 59.2% but introduced 85 new False Positives in our sample. These fall almost entirely into low toxicity bins (0–4), with 28 of the 85 cases in bin 1 and 24 in bin 2, indicating that the fine-tuned model over-triggers on content with low but non-zero annotated toxicity. Qualitative inspection reveals two principal causes. The first is domain-specific vocabulary that co-occurs with toxic content during training but is benign in context: military jargon (*"corporal can you dick some private soldiers"*), political metaphor (*"colonialist capitalist mad-libs"*), and colloquial references to audience behavior on Bluesky (*"disposed of all the annoying bluesky elders"*). The second cause is short posts whose brevity removes disambiguating context, leaving the model to over-rely on surface vocabulary. Posts such as *"marxists leninists hate this one trick"* (ft prob: 0.90) or *"When life hands you lemons Joe Biden kills you with them"* (ft prob: 0.77) illustrate cases where political framing or colloquial phrasing triggers above-threshold probabilities despite no genuine toxic content being present.

Taken together, these patterns are consistent with the interpretation that fine-tuning on cross-platform data broadens the model's detection vocabulary but simultaneously reduces its calibration on borderline or context-dependent instances, a well-documented trade-off in domain adaptation for classification tasks.

6.2 Validation versus Test Generalization Gap

The fine-tuned model achieves F1=70.2% on the in-domain validation set (Reddit + Koo) but 66.7% on the out-of-domain test set (Bluesky + Voat), a gap of 3.5 points. The test platforms exhibit a substantially different character-length distribution (Figure 7), particularly Bluesky's short-post format. Given that short posts disproportionately appear in the FP-New category (Section 6.1), this distributional mismatch is likely a contributing factor to the generalization gap beyond any residual platform-specific linguistic fingerprints.

6.3 Ablation: Effect of Length Balancing

To assess the contribution of the length-balancing component of our sampler, we ran a second experiment using a sampler that balances by both toxicity label *and* character length (Experiment 2). Table 4 compares the two configurations on the test set.

Table 4: Ablation: Effect of adding character-length balancing (test set).

Sampler	Acc.	Prec.	Rec.	F1
Tox-balanced only (Exp. 1, ours)	**62.8%**	**61.2%**	**73.2%**	**66.7%**
Tox + Length balanced (Exp. 2)	62.7%	61.1%	71.8%	66.0%

Both samplers balance toxic and non-toxic labels across 10 toxicity bins. The difference is that Experiment 2 additionally stratifies within each toxicity bin by character length (5 bins), creating a 10×5 joint stratum. Contrary to our hypothesis, the toxicity-only sampler (Experiment 1) achieves marginally better test performance: +1.4 points Recall and +0.7 points F1. A notable difference is in the validation-to-test gap: Experiment 1 shows a 3.5-point gap (Val F1=70.2%, Test F1=66.7%), while Experiment 2 shows only a 0.3-point gap (Val F1=66.3%, Test F1=66.0%). This suggests that length-balancing produces a validation estimate more representative of test distribution, but at the cost of slightly lower absolute performance, possibly because the additional constraint reduces the effective training signal per toxicity class.

6.4 Challenges and Limitations

Emoji-encoded Algospeak. Our preprocessing pipeline strips all non-textual characters and emojis. This excludes visual-semantic Algospeak (e.g., using leaf or scissors emojis as coded symbols), a growing evasion vector documented in recent moderation literature [4]. Future work should explore multimodal preprocessing that retains emoji semantics as token features.

Temporal drift. Both MADOC and the Algospeak lexicon reflect a specific time window. Newly coined evasion terms (post-2024 slang, platform-specific neologisms) are not covered by the current model. Periodic re-fine-tuning on fresh data would be required for production deployment.

Balanced validation set. The validation set was artificially balanced to 50/50 toxic/non-toxic via stratified sampling. The reported val F1 of 70.2% was therefore measured under controlled conditions that may overstate performance relative to a fully naturalistic class distribution, and should be interpreted accordingly.

7 Conclusion

We presented a toxicity-balanced fine-tuning approach for detecting Algospeak in cross-platform social media content. By stratifying training samples across fine-grained toxicity bins and fine-tuning a DistilBERT model on the MADOC dataset [3], we achieved substantial improvements in Recall (+40.0 points) and F1-Score (+21.6 points) over an untuned baseline on an out-of-domain test set. An ablation study further showed that adding character-length balancing does not improve test performance, suggesting that toxicity-bin stratification alone is the key factor. These results underscore the importance of domain-aware data preparation for detecting evolved toxic language. The methodology is lightweight, reproducible, and extensible to other platforms or evasion typologies.

References

[1] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.

[2] Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.

[3] Fillies, S., & Paschke, A. (2024). *The MADOC Dataset: Multi-Platform Aggregated Dataset of Online Communities*. Zenodo. `https://doi.org/10.5281/zenodo.14637314`.

[4] Fillies, J., & Paschke, A. (2024). Simple LLM based approach to counter Algospeak. In *Proceedings of WOAH 2024*.

[5] Lu, J., et al. (2023). Facilitating fine-grained detection of Chinese toxic language. In *Proceedings of ACL 2023*.

[6] Davidson, T., Warmsley, D., Macy, M., & Weber, I. (2017). Automated hate speech detection and the problem of offensive language. In *Proceedings of ICWSM 2017*.

[7] Wiegand, M., Ruppenhofer, J., & Eder, E. (2021). Implicitly abusive language—what does it actually look like and why are we not getting there? In *Proceedings of NAACL-HLT 2021*, pages 576–587.

A Appendix: Supplemental EDA Figures

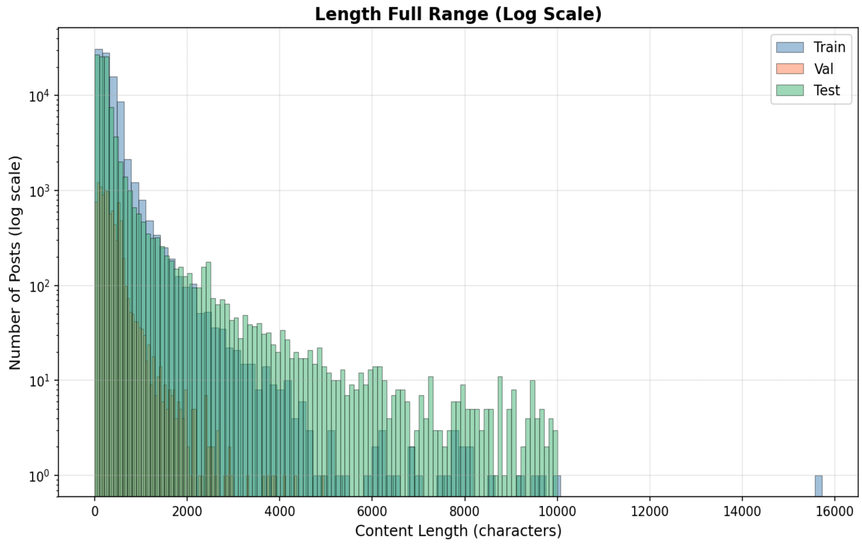

Figure 7: Post length distribution (full range, log scale) across Train, Validation, and Test splits. Train (Reddit+Koo) and Test (Bluesky+Voat) show markedly different length profiles, reflecting platform-specific writing norms (Bluesky's short-post format vs. Reddit's longer discourse). This distributional mismatch contributes to the validation-to-test generalization gap discussed in Section 6.3.

B In-Domain Confusion Matrices

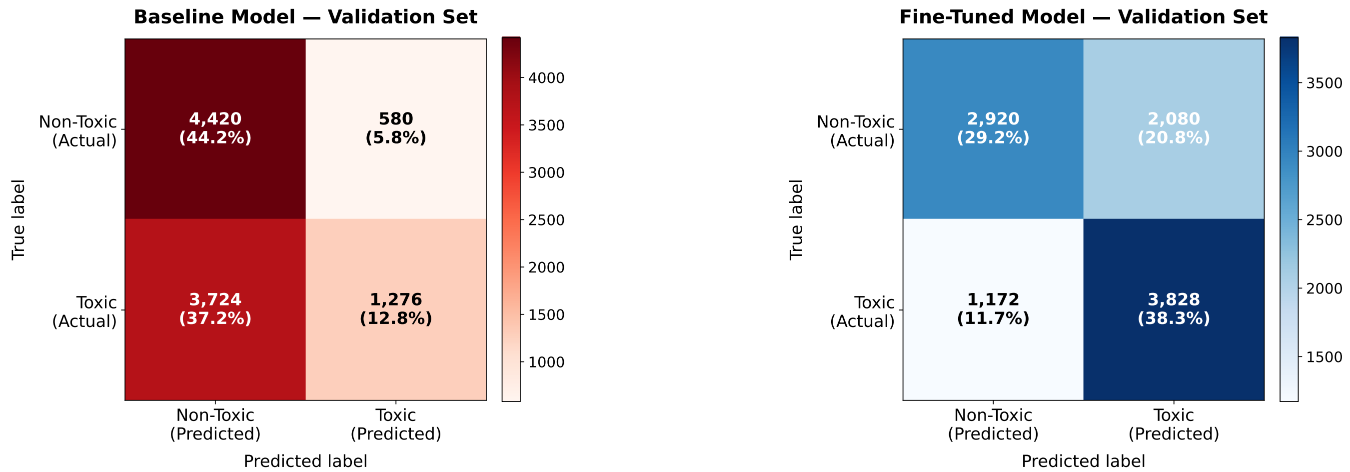

Figure 8: Confusion matrices on the in-domain validation set. Left: Baseline. Right: Fine-tuned. The validation set is balanced 50/50 toxic/non-toxic.

C AI Usage Disclosure and Reflection

Disclosure: Generative AI (Claude by Anthropic, Gemini by Google) was used in the following ways: (1)brainstorming research directions (2) debugging Python code implementation issues (3) assisting with LaTeX formatting and section structure; (4) suggesting phrasings and improving the clarity of written sections. All experimental results, code, figures, and scientific claims are original and produced by the authors.

Reflection: AI tools substantially accelerated LaTeX formatting and helped maintain a consistent academic register throughout the paper. However, human judgment was essential for interpreting experimental results, identifying the correct citations, and ensuring the research framing accurately reflected the limitations of our methodology.